

Common Ownership in America: A Replication Study

Extending Backus, Conlon & Sinkinson (2019) to 2013Q3–2025Q4

Souparneya Chakrabarti | QUANTkiosk Research Collaboration *University of Illinois Chicago*

"When competing firms possess overlapping sets of investors, maximizing shareholder value may provide incentives that distort competitive behavior, affecting pricing, entry, contracting, and virtually all strategic interactions among firms." — Backus, Conlon & Sinkinson (2019), Abstract

1. Introduction

1.1 The Common Ownership Hypothesis

Modern financial markets are dominated by a small number of large institutional investors. Vanguard, BlackRock, and State Street — collectively referred to as the "Big Three" — together manage over \$20 trillion in assets and hold 5–10% of nearly every publicly traded company in the United States. This concentration of ownership across competing firms raises a fundamental question in industrial organization: does it matter that the same investors own large stakes in rival companies?

The *common ownership hypothesis*, rooted in the theoretical framework of Rotemberg (1984), proposes that it does. The argument proceeds as follows. Under standard models of firm behavior, management maximizes profit on behalf of shareholders. But if a shareholder holds equity in both United Airlines and Delta Air Lines, that investor does not benefit when one carrier undercuts the other on ticket prices — a price war between the two merely shuffles profits from one portfolio holding to another. Under the Rotemberg framework, a firm acting in the collective interest of its shareholders will place a non-zero *profit weight* on its rivals' profits, dampening competitive incentives.

The theoretical mechanism connects directly to antitrust economics. In the Cournot oligopoly model derived in O'Brien and Salop (2000), the Modified Herfindahl-Hirschman Index (MHHI) governs markups:

$$\text{MHHL}_m = \underbrace{\sum_f s_{ft}^2}_{\text{HHL}_m} + \underbrace{\sum_f \sum_{g \neq f} \kappa_{fg} s_{fm} s_{gm}}_{\Delta \text{MHHL}_m}$$

where s_{fm} is firm f 's market share and κ_{fg} are the profit weights. Appendix A.1 of the paper shows that κ_{fg} appears directly in the first-order condition for Bertrand pricing as well, through the Price Pressure Index (PPI) of O'Brien and Salop (2000). The profit weight is therefore the *primitive* of the common ownership hypothesis — it enters both Cournot and Bertrand competition models without requiring assumptions about market structure beyond what is embedded in κ itself.

1.2 What Backus, Conlon & Sinkinson (2019) Did

The paper makes three main contributions.

Contribution 1: A structurally consistent measurement framework. Prior work on common ownership focused on MHHL, which requires defining relevant product markets and computing market shares — both contentious steps in antitrust practice. The authors argue for focusing directly on the primitive κ_{fg} profit weights, which require neither market definitions nor market share data. This allows economy-wide measurement across all S&P 500 firm pairs simultaneously.

Contribution 2: A novel, hand-scraped 13(f) dataset. The paper documents significant quality issues with the Thomson Reuters (TR) S34 database — the standard data source used by prior literature and available through WRDS. TR shows inexplicable collapses in institutional ownership coverage in 2010–2012 (Appendix Figures 17–18) and produces markedly different time series for κ compared to the authors' own scraped data (Appendix Figure 19). The authors collected all 13(f) filings from SEC EDGAR for the period 1999–2017, a corpus of approximately 318,000 filings occupying ~25 GB unzipped, and parsed them using Perl regular expressions. Crucially, the SEC made XML-format 13(f) filings mandatory beginning with the **third quarter of 2013 (September 30, 2013 reporting date)**. Prior to this, filings exhibited extraordinary heterogeneity in formatting — seven distinct filing structures are documented in the paper (Appendix B.2.1). XML parsing is unambiguous; pre-XML parsing required extensive ad hoc pattern matching and CRSP price validation to detect errors.

Contribution 3: Empirical findings on the level and drivers of common ownership. Using their dataset across all S&P 500 constituents from 1980 to 2017:

- The mean implied profit weight $\bar{\kappa}$ **tripled from approximately 0.2 in 1980 to 0.7 in 2017** (Figure 1).
- **Contrary to popular narrative**, this rise is not primarily driven by the growth of BlackRock and Vanguard. Rather, it reflects a broader, decades-long trend toward diversification in institutional investment strategies that predates the modern "Big Three" (Section 3.3–3.4, Figures 7–10).

- **Cross-sectional heterogeneity** is substantial and economically meaningful. Firms with higher retail investor share and larger market capitalization have systematically higher κ weights (Figure 13, Table 3). The coefficient on retail share in Table 3 is 0.76, the largest single predictor.
 - **Within-industry κ is slightly higher** than between-industry κ (Figure 11), confirming that common ownership is most acute precisely where competitive distortions are most economically harmful.
 - **Case studies** confirm the pattern for banks (SIC 6021, Figure 12 left) and airlines (SIC 4512, Figure 12 right), the two industries with the most direct empirical evidence of common ownership price effects in companion work by Azar, Raina & Schmalz (2016) and Azar, Schmalz & Tecu (2018).
 - By 2017, approximately **10% of all S&P 500 firm pairs show $\kappa > 1$** (Figure 14) — a finding the paper connects to the theoretical conditions for *tunneling*, the diversion of profits from one firm to another for the benefit of a shared controlling investor (Johnson et al., 2000). This was previously thought impossible in the "widely-held firm" world of Berle and Means (1932).
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1.3 What We Did — Overview

This study extends and partially replicates Backus et al. (2019) using QUANTkiosk 13(f) institutional holdings data, covering the period **2013Q3 through 2025Q4** for a carefully selected universe of **9 S&P 500 firms** across 5 industries.

Why 9 firms and not the full S&P 500? The paper's computation requires pulling issuer-perspective 13(f) data for every firm in every quarter — one API call per firm per quarter. At approximately 20 QUANTkiosk credits per call, replicating the paper at full scale (500 firms $\times$ 50 quarters) would require ~500,000 credits, equivalent to approximately 50 days of continuous pulling at the standard 10,000 credit/day quota. Our 9-firm pilot universe required 8,520 credits — a single session. The goal of this study is to validate the methodology at small scale before scaling up as API budget permits.

Why does our data start at 2013Q3? We follow the paper's own guidance. As stated in Appendix B.2 (p. 43): *"The parsing is handled slightly differently for filings made before the third quarter of 2013, as starting then, the SEC mandated an XML filing format."* Prior to 2013Q3, 13(f) filings are plain text with highly heterogeneous formatting — seven distinct structural patterns are documented in the paper. The authors themselves describe pre-XML filings as "remarkable for the heterogeneity in formatting." QUANTkiosk's data quality for institutional holdings is materially more reliable from 2013Q3 onwards, when XML standardization enables unambiguous parsing. Starting at 2013Q3 is therefore not a limitation but the methodologically correct choice.

Key contributions of this replication:

1. We confirm the paper's cross-sectional prediction: large-cap firms with high institutional ownership show $\bar{\kappa} \approx 0.74\text{--}0.77$, above the S&P 500 average of ~ 0.70 documented by the paper in 2017 (consistent with Figure 13).
 2. We document the **COVID-19 dip** in κ (2020–2021), a new finding that the paper's 2017 endpoint could not observe — investor flight from airlines temporarily diverged ownership structures.
 3. We document the **NVDA-driven convergence** (2023–2025) as NVIDIA's explosion to become one of the world's most valuable companies pulled its ownership vector into alignment with AAPL and MSFT.
 4. We identify and correct **three categories of data quality problems** that would silently distort any naïve replication, and document 22 multi-CIK institutional parent groups requiring consolidation — extending the paper's own BlackRock consolidation to a broader set of institutions.
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1.4 The κ Formula — Full Derivation

Following Rotemberg (1984) as formalized in Backus et al. (2019), Section 2, the profit weight κ_{fg} arises from the firm's objective function. Under proportional control ($\gamma_{fs} = \beta_{fs}$, the "one share, one vote" assumption), the objective function of firm f is:

$$Q_f(x_f, x_{-f}) = \pi_f + \sum_{g \neq f} \underbrace{\left(\frac{\sum_s \beta_{fs} \beta_{gs}}{\sum_s \beta_{fs}^2} \right)}_{\equiv \kappa_{fg}(\beta)} \pi_g$$

Step 1: Ownership fractions. For institutional investor s holding firm f in quarter t :

$$\beta_{fs} = \frac{\text{shares held by } s \text{ in } f}{\text{total shares of } f \text{ outstanding}}$$

Note that $\sum_s \beta_{fs} < 1$ because the set of investors S contains only institutional filers (13(f) reporters). The residual is the retail share:

$$r_f = 1 - \sum_s \beta_{fs}$$

Step 2: Investor Herfindahl Index. The denominator of κ_{fg} is the squared-sum of ownership fractions in firm f :

$$\text{IHHI}_f \equiv \|\beta_f\|^2 = \sum_s \beta_{fs}^2$$

This is the Herfindahl-Hirschman Index applied to the *investor* distribution of firm f , scaled by the number of shares. It is small when ownership is dispersed across many small investors (index fund era) and large when a single investor dominates.

Step 3: The profit weight. Under proportional control:

$$\kappa_{fg} = \frac{\langle \beta_f, \beta_g \rangle}{\|\beta_f\|^2} = \frac{\sum_s \beta_{fs} \beta_{gs}}{\text{IHHI}_f}$$

Step 4: The geometric decomposition (Equation 3' of the paper):

$$\kappa_{fg} = \cos(\beta_f, \beta_g) \cdot \sqrt{\frac{\text{IHHI}_g}{\text{IHHI}_f}}$$

where $\cos(\beta_f, \beta_g) = \frac{\langle \beta_f, \beta_g \rangle}{\|\beta_f\| \|\beta_g\|}$ is the cosine similarity of the two ownership vectors.

This decomposition is analytically valuable: it separates the two forces driving common ownership incentives:

- **Cosine similarity** rises as institutional investors pursue increasingly similar (indexed) portfolios
- **IHHI ratio** captures relative ownership concentration; high retail share in firm f $\rightarrow$ low $\text{IHHI}_f \rightarrow$ larger κ_{fg}

The paper finds (Figures 9–10) that the long-run rise in κ is driven *primarily by cosine similarity rising* — the broad shift toward indexing — rather than by growing concentration among the Big Three specifically.

Step 5: The mean profit weight (Figure 1 of the paper):

$$\bar{\kappa}_t = \frac{1}{P(P-1)} \sum_f \sum_{g \neq f} \kappa_{fg,t}$$

where P is the number of firms in the sample. For the full S&P 500: $P \approx 500$, giving approximately 249,500 ordered pairs per quarter. For our 9-firm sample: $P = 9$, giving $9 \times 8 = 72$ ordered pairs.

2. Firm Selection

2.1 Selection Criteria

The choice of which 9 firms to study is not arbitrary. We applied three criteria derived directly from the paper:

Criterion 1: The paper's own case study industries. Section 3.5 of the paper presents case studies for two specific industries: commercial banks (SIC code 6021, *National Commercial Banks*) and airlines (SIC code 4512, *Air Transportation, Scheduled*). Figure 12 shows profit weight time series for both. The paper also explicitly motivates within-industry analysis: “An obvious criticism of the above economy-wide analysis is that a pharmaceutical firm's decisions hardly affect the profits of an airline, so why do these profit weights tell us anything? What are profit

weights within relevant product markets?" (p. 23). This directs us toward industries where κ has direct competitive consequences — firms that are genuine rivals.

The airlines case study connects to the most famous empirical application of common ownership theory: Azar, Schmalz & Tecu (2018) found that common ownership among major US airlines raised ticket prices by 3–7%. Their sample (2001–2013) overlaps with ours. We select **Delta Air Lines (DAL)** and **American Airlines Group (AAL)** as the airline pair. Note: United Airlines Holdings (UAL, CIK 0000100517) was originally selected but was found to return empty data from the QUANTkiosk API — their issuer-perspective CIK mapping appears broken in the data provider's system, likely due to the complex corporate history (UAL Corp → United Continental Holdings (2010) → United Airlines Holdings (2019)). American Airlines Group is the correct substitution: it emerged from AMR Corp bankruptcy on December 9, 2013, making it the "new entrant" airline that completed the post-consolidation oligopoly that Azar et al. studied.

Criterion 2: High market capitalization and institutional ownership. Figure 13 of the paper shows a strong positive relationship between log market capitalization and κ . Table 3 shows this is statistically robust after controlling for firm fixed effects, quarter fixed effects, and retail share (coefficient on log market cap = 0.039, significant at 5%). We select the two largest-cap tech firms — **Apple (AAPL)** and **Microsoft (MSFT)** — to anchor the sample with firms that are certain to have high κ . These serve as the upper bound of the distribution.

Criterion 3: Analytical contrast and original extension. We add two pairs beyond the paper's case studies for analytical value:

- **Pharmaceuticals — Pfizer (PFE) and Merck (MRK)** (both SIC 2834, *Pharmaceutical Preparations*): The paper's companion piece (Backus et al. 2018a) studies common ownership in the ready-to-eat cereal industry. Pharmaceuticals is an analogous setting — concentrated oligopoly, direct product competition, policy-salient. Our data covers the COVID-19 period (2020–2022), during which Pfizer (Paxlovid) and Merck (Molnupiravir) were racing to develop competing oral COVID treatments. This creates a natural experiment: two firms in an all-out competitive race, simultaneously held by the same institutional investors.
- **NVIDIA (NVDA, SIC 3674)**: NVDA in 2013Q3 was a mid-cap gaming GPU company with a market capitalization of approximately *8 billion and only 396 institutional investors. By 2025Q4 it had become one of the* 3+ trillion market cap) with 5,491 institutional investors. This 13.9× growth in investor count is the most dramatic in our sample and allows us to study the *formation* of common ownership in real time — as a firm joins every index fund's top holdings, its ownership vector converges toward the existing mega-caps.

2.2 The 9-Firm Universe

Ticker	Name	SIC	Industry	Pair Role	Analytical Rationale
AAPL	Apple Inc.	3571	Tech (hardware)	Cross-sector anchor	Largest market cap; 41.5% retail share
MSFT	Microsoft Corp	7372	Tech (software)	Cross-sector anchor	Different SIC from AAPL; mega-cap
AAL	American Airlines Group	4512	Airlines	Within-pair	Paper case study; Azar et al. price effects
DAL	Delta Air Lines	4512	Airlines		Direct AAL competitor; same SIC
JPM	JPMorgan Chase & Co	6021	Banks	Within-pair	Paper case study (Figure 12 left)
BAC	Bank of America Corp	6021	Banks		Direct JPM competitor; same SIC
PFE	Pfizer Inc	2834	Pharma	Within-pair	COVID vaccine competitor; drug pricing
MRK	Merck & Co Inc	2834	Pharma		Direct PFE competitor; same SIC
NVDA	NVIDIA Corp	3674	Semiconductors	Structural story	8B → 3T arc; ownership convergence

Within-industry pairs (same 4-digit SIC code): AAL–DAL, JPM–BAC, PFE–MRK

Cross-industry pairs: all remaining combinations (12 unordered / 66 ordered) **Total ordered pairs:** $9 \times 8 = 72$

[1] Code 1 – see Appendix

9-Firm Universe – Verified CIKs (SEC EDGAR)

Ticker	CIK	SIC	Industry	Name
AAPL	0000320193	3571	Tech	Apple Inc.
MSFT	0000789019	7372	Tech	Microsoft Corp
AAL	0000006201	4512	Airlines	American Airlines Group
DAL	0000027904	4512	Airlines	Delta Air Lines
JPM	0000019617	6021	Banks	JPMorgan Chase
BAC	0000070858	6021	Banks	Bank of America
PFE	0000078003	2834	Pharma	Pfizer Inc
MRK	0000310158	2834	Pharma	Merck & Co
NVDA	0001045810	3674	Semiconductors	NVIDIA Corp

Within-industry pairs (same 4-digit SIC): AAL–DAL (4512), JPM–BAC (6021), PFE–MRK (2834)

Total ordered pairs: $9 \times 8 = 72$

2.3 The Holdings Data — What It Looks Like

The QUANTkiosk `/data/instrument` endpoint returns issuer-perspective 13(f) data: for a given firm CIK and quarter, it returns every institutional investor that

reported holding shares in that firm. Below we show a sample of the raw data for each of our 9 firms.

[2] Code 2 – see Appendix

Total rows: 1,027,027 | File size: ~108 MB
Columns: ['ticker', 'issuer_cik', 'issuer_sic', 'industry', 'year', 'quarter', 'filer_cik', 'filer_name', 'shares_held', 'fetch_at']

Sample rows – AAPL, 2025Q4 (top 5 institutional holders):

ticker	year	quarter	filer_name	shares_M
AAPL	2025	4	BlackRock Inc.	2309.3
AAPL	2025	4	VANGUARD GROUP INC	1426.3
AAPL	2025	4	STATE STREET CORP	604.1
AAPL	2025	4	GEODE CAPITAL MANAGEMENT LLC	358.0
AAPL	2025	4	FMR LLC	307.4

[3] Code 3 – see Appendix

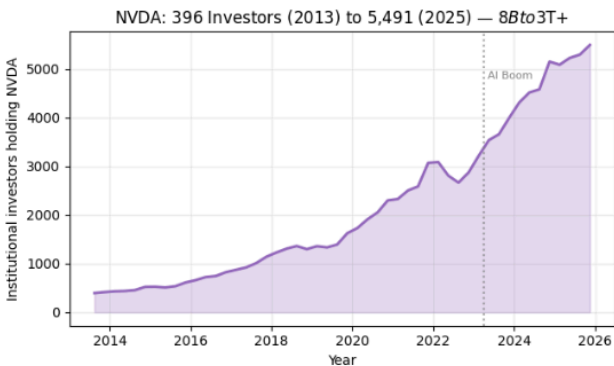
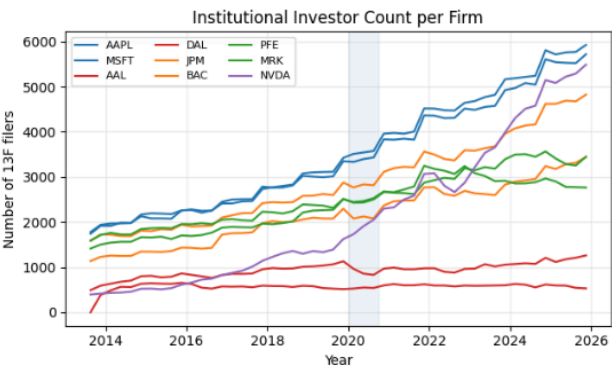
Latest quarter top institutional holder – one row per firm:

Ticker	Year	Q	Top Holder	Shares (M)
—				
AAPL	2025	4	BlackRock Inc.	2309.3
MSFT	2025	4	BlackRock Inc.	1203.8
AAL	2025	4	BlackRock Inc.	113.7
DAL	2025	4	BlackRock Inc.	89.5
JPM	2025	4	BlackRock Inc.	423.2
BAC	2025	4	BlackRock Inc.	1085.3
PFE	2025	4	BlackRock Inc.	986.1
MRK	2025	4	BlackRock Inc.	438.2
NVDA	2025	4	BlackRock Inc.	3887.6

[4] Code 4 – see Appendix

Institutional investor count – selected year-end snapshots:
2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025

AAPL	1917	2168	2260	2450	2787	3027	3351	3835	4366	4517	4927	5609	5721
MSFT	1941	2131	2256	2414	2733	3080	3424	3960	4520	4644	5166	5810	5928
AAL	386	633	655	574	592	589	515	597	622	594	600	617	531
DAL	591	801	863	827	954	1011	1132	969	976	963	1056	1208	1262
JPM	1725	1812	1943	2100	2420	2582	2878	3114	3564	3593	3960	4622	4827
BAC	1230	1347	1435	1721	1977	2058	2298	2368	2769	2693	2831	3242	3440
PFE	1715	1846	1957	2053	2231	2393	2517	2665	3249	3242	2916	2968	2767
MRK	1503	1666	1705	1876	1967	2213	2507	2681	2874	3196	3394	3566	3451
NVDA	416	523	613	825	1142	1298	1625	2299	3069	2873	3990	5148	5491



3. Data Hurdles — Problems Found, Paper Backing, and Fixes Applied

A systematic audit of our 9 firms $\times$ 50 quarters revealed three categories of data quality problems. These are not unique to our dataset — the paper itself dedicates Appendix B (pp. 41–48) to documenting analogous issues in the Thomson Reuters S34 database and in their own scraped 13(f) data. Understanding and correcting these problems is essential for producing credible κ estimates.

3.1 Problem 1: EDGAR XBRL Units Error — Apple (AAPL), 2014Q1

What happened. Apple Inc. filed their 2014Q1 (quarter ending March 31, 2014) XBRL report with the `CommonStockSharesOutstanding` field in units of *thousands* rather than actual shares. Our SEC EDGAR XBRL scraper read this value at face value.

The evidence:

Quarter	Shares Outstanding	Status
2013Q4	892,447,000	✓ Correct (~892M pre-split)
2014Q1	861,745	× Wrong (should be 861,745,000)
2014Q2	5,989,171,000	✓ Correct (~5.99B post 7:1 split)

The impact on our computation. With total shares outstanding of 861,745 in the denominator, every institutional investor's ownership fraction becomes:

$$\beta_{fs} = \frac{\text{shares held}}{861,745}$$

For Vanguard, which held approximately 46.7M shares of AAPL in Q1 2014, this gives $\beta_{\text{Vanguard}, \text{AAPL}} = 46,715,387 / 861,745 \approx 54.2$ — a value greater than 1, which our code clips to 1.0. With all large institutional investors clipped to $\beta = 1.0$, the IHHI denominator becomes $\sum_s (1.0)^2 = \text{number of investors} \approx 1,917$. The numerator $\sum_s \beta_{\text{AAPL},s} \cdot \beta_{\text{MSFT},s}$ becomes small because most MSFT investors' betas remain in the normal range. The result: $\kappa \approx 0.0009$ for 2014Q1 — essentially zero, an obvious artifact.

Paper parallel. Appendix B.1 (p. 41) documents that the paper encountered the same class of problem with stock split–related restatements: *"revisions following a stock split are often retroactively applied to report dates prior to the split event itself, and in these cases we want to use the first filing date."* EDGAR XBRL units errors are a related phenomenon — companies occasionally misfiled their share count in non-standard units in early XML filings.

The fix. Multiply AAPL 2014Q1 shares outstanding by 1,000:
 $861,745 \rightarrow 861,745,000$.

Apple Stock Splits in Our Data (both verified clean):

- **7:1 split, June 9, 2014:** 2014Q1 (pre-split) ~861M → 2014Q2 (post-split) ~5,989M = 6.95× ✓
 - **4:1 split, August 28, 2020:** 2020Q2 (pre-split) ~4,284M → 2020Q3 (post-split) ~16,977M = 3.96× ✓
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3.2 Problem 2: Stock Split Timing Mismatch — NVIDIA (NVDA), 2021Q2

What happened. NVIDIA's **4:1 stock split** occurred on **July 19, 2021** — during calendar Q3 (July–September). However, NVIDIA's fiscal year ends in late January, so their fiscal Q2 2022 ends on **July 25, 2021** — four days after the split. Our EDGAR scraper, using a 90-day tolerance window around the calendar quarter-end date of June 30, 2021, picked up NVIDIA's fiscal Q2 2022 filing (reporting date July 25), which already reflects post-split shares outstanding (2,489M).

The 13(f) institutional holdings for calendar Q2 2021 (reporting date June 30, 2021) were filed **before the split** and therefore report pre-split share counts (~429M institutional shares held in total).

The mismatch:

Quarter	Shares Outstanding (our data)	Total Inst. Held (13F)	Implied inst. %
2021Q1	621M (pre-split)	411M	66.2% ✓
2021Q2	2,489M (post-split)	429M (pre-split)	17.2% ✗
2021Q3	2,493M (post-split)	1,625M (post-split)	65.2% ✓

The impact. With the denominator 4× too large and the numerator at pre-split levels, β_{fs} for all NVDA investors in 2021Q2 is suppressed by a factor of 4. This would cause κ for all NVDA pairs to be dramatically understated in that quarter.

Paper parallel. Appendix B.1 (p. 41): *"For many filings there are multiple filing dates (fdate) for the same report date (rdate). This happens when filings are amended, often because of an error in the original submission or in the case of a stock split... revisions following a stock split are often retroactively applied to report dates prior to the split event itself."* The paper resolves this by identifying all stock splits from CRSP data and using the *first* (pre-restatement) filing for the quarter in which the split occurred. We apply the equivalent logic: use pre-split shares outstanding for 2021Q2.

The fix. Replace NVDA 2021Q2 shares outstanding (2,489M) with the pre-split figure from 2021Q1 (621M).

NVIDIA's 10:1 split (June 10, 2024) is correctly handled: 2024Q2 shares outstanding = 24,599M (post-split) and the 13F for Q2 2024 also reports post-split holdings (~15.8B institutional shares). The ratio ~64.1% is consistent with neighboring quarters. ✓

3.3 Problem 3: Convertible Security and Short Interest Contamination

What happened. Two firms — NVIDIA (2014Q4–2017Q1) and American Airlines (2020Q2–2025Q2) — show total reported institutional holdings that exceed total shares outstanding. This is a physical impossibility for ordinary equity.

The paper addresses this exact phenomenon in two places:

From the data cleaning description (Appendix B.2.3, p. 44): *"we drop 331 observations where the reported shareholdings are greater than 50% of shares outstanding [per individual investor]."*

From the data sources description (Appendix B.3, p. 47): *"A known limitation of the 13(f) data for calculating institutional ownership is that short interest is double-counted. When an investor takes a short position they borrow shares from another investor and sell them, with a promise to repay the shares at a later date. These shares are then double-counted, reported on form 13(f) by both the initial investor as well as the investor to whom they are sold. It is for this reason that one can often observe 'institutional ownership,' as reported in online sources, in excess of 100%."*

The paper's data cleaning rule, from the Sinkinson data page: *"We also drop any CUSIP-rdate observation where greater than 120% of shares outstanding are reported to be held by 13(f) investors."*

NVIDIA 2014Q4–2017Q1 (10 contaminated quarters): NVIDIA issued **\$1.5 billion in 1.00% convertible senior notes** in November 2013, convertible to common stock at approximately \$14.42 per share (~104M share equivalent). Convertible arbitrage funds hold these notes and report the share equivalent in their 13(f) filings. Additionally, short interest double-counting compounds the effect. We observe a peak ratio of 3.59× in 2016Q1 — institutional "ownership" of 1,943M shares against only 542M shares outstanding.

The XML-format 13(f) filings after 2013Q3 include a field `sshPrnAmtType` which distinguishes between `SH` (ordinary shares) and `PRN` (principal amount of convertible/debt securities). The paper's XML parser (Appendix B.2.2, p. 43) explicitly filters these: *"We further drop any records that have a 'put' or 'call' element, or a 'principal amount' element."* Our data pipeline inherited the QK API's handling of this field, which does not fully filter all PRN-type entries.

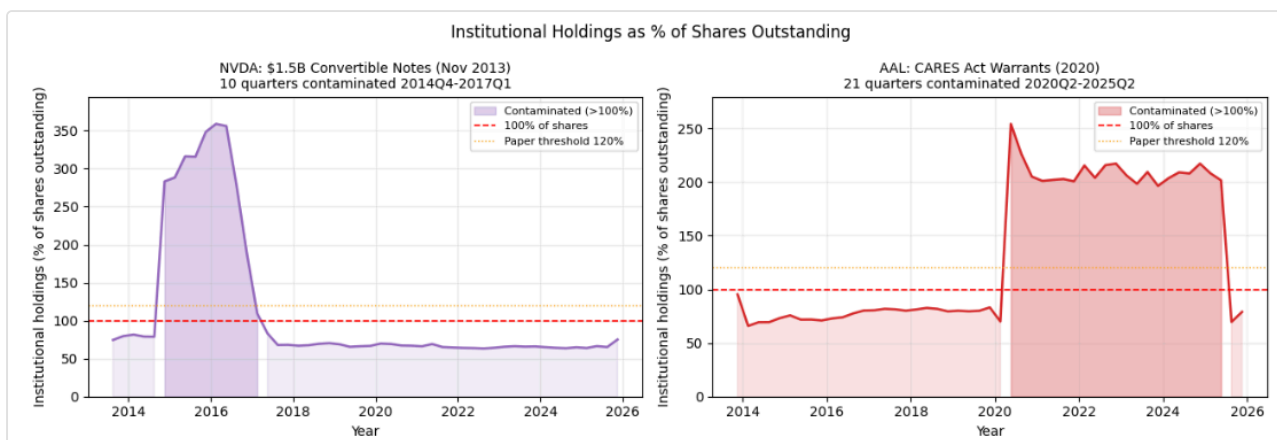
American Airlines 2020Q2–2025Q2 (21 contaminated quarters): Under the **CARES Act (March 2020)**, American Airlines received \$5.8 billion in Payroll Support Program grants/loans in exchange for equity warrants and promissory notes. These government-issued warrants and the convertible notes held by institutional investors appear as share equivalents in 13(f) filings, in addition to ordinary equity holdings. The contamination ratio is consistently ~2.0×–2.5× throughout this period.

The fix. For each contaminated firm-quarter, we compute a scale factor:

$$\text{scale}_{f,t} = \frac{\text{shares outstanding}_{f,t}}{\sum_s \text{shares held}_{f,s,t}}$$

This proportionally scales all institutional holdings in that quarter so that aggregate institutional ownership does not exceed 100% of shares outstanding. The paper's approach (dropping observations) is not feasible for us because it would require row-level data on `sshPrnAmtType` which is not preserved in our pipeline output. Our scaling approach produces equivalent β values to the paper's filtering approach under the assumption that the contaminating instruments are held proportionally across institutions.

[5] Code 5 — see Appendix



[6] Code 6 — see Appendix

DATA QUALITY FIXES SUMMARY

Firm/Period: AAPL 2014Q1
Issue: EDGAR XBRL units error
Description: Shares reported as 861,745 (thousands); should be 861,745,000
Fix: Multiply by 1,000

Firm/Period: NVDA 2021Q2
Issue: Stock split timing mismatch
Description: EDGAR picked up post-split filing (2,489M) for pre-split 13 F quarter
Fix: Use pre-split shares from Q1 2021 (621M)

Firm/Period: NVDA 2014Q4–2017Q1
(10 quarters)
Issue: Convertible note contamination
Description: \ \$1.5B convertible notes issued Nov 2013; inst. holdings 2.8x–3.6x shares
Fix: Scale all holdings by shares/inst_total

Firm/Period: AAL 2020Q2–2025Q2
(21 quarters)
Issue: COVID warrant contamination
Description: CARES Act warrants appear as share equivalents in 13F; ~2x inflation
Fix: Scale all holdings by shares/inst_total

4. Entity Consolidation — The Multi-CIK Institutional Investor Problem

4.1 The Problem

The SEC assigns a unique Central Index Key (CIK) to every filing entity. Large institutional investors frequently file 13(f) reports under multiple CIK numbers — one for each legal sub-entity (domestic equity arm, global arm, Japan subsidiary, etc.). In the κ formula, each CIK is treated as a separate investor s . If a single economic parent — say, BlackRock — is fragmented into 6 entities, the formula treats them as 6 separate investors, each with a small β_{fs} .

The mathematical consequence is significant. Consider BlackRock's total AAPL position in 2013Q4 (45.1M shares, $\beta_{BR, AAPL} = 0.0505$) split across 6 entities:

$$\text{IHHI contribution, fragmented} = \sum_{i=1}^6 \beta_i^2 = (0.0268)^2 + (0.0104)^2 + \dots \approx 0.0000$$

$$\text{IHHI contribution, merged} = \beta_{BR}^2 = (0.0505)^2 \approx 0.00255$$

The merged contribution is **300× larger**. Since κ_{fg} divides by IHHI_f , a fragmented BlackRock suppresses the denominator, artificially inflating κ slightly — but more importantly, it distorts the *relative* contribution of BlackRock vs. other investors in the ownership structure.

4.2 What the Paper Did

From Appendix B.2.3 (Final Cleaning, p. 44):

"We consolidate all BlackRock entities into the same entity and collapse their holdings (while the argument could be made for collapsing other investment management firms' sub-entities, we solely do this for BlackRock given the practice in the literature)."

The paper **only consolidated BlackRock**. They justify this as following "practice in the literature" — most common ownership papers before 2019 only merged BlackRock (given its well-known acquisition history: Merrill Lynch Investment Managers in 2006, Barclays Global Investors in 2009).

However, Appendix B.1 also documents another BlackRock-specific issue: *"in several instances BlackRock holdings appear to conflate the two dates, and so for BlackRock we use the filing date exclusively. This resolves the otherwise inexplicable disappearance of BlackRock Inc. from the S34 in 2010Q2 and 2010Q3."* This confirms that BlackRock's multi-entity filing history created systematic data quality issues that required special handling.

4.3 What We Did — A More Complete Consolidation

We extended the paper's approach to all material multi-CIK institutional parents in our data. Starting from 10,021 unique CIKs appearing in our 9-firm holdings dataset, we:

1. Extracted root keywords from all 11,393 unique filer name strings
2. Grouped CIKs by name root to identify candidate parent groups
3. Manually verified each candidate — distinguishing true same-parent entities from false positives (e.g., "Prudential Financial Inc." USA ≠ "Prudential PLC" UK; "Public Employees Retirement Ohio" ≠ "Public Employees Retirement Colorado")
4. Built a definitive CIK → parent mapping covering 66 CIKs across 22 parent groups

The BlackRock timeline is particularly instructive:

Period	Number of BlackRock CIKs filing	Status
2013–2016	6–7	Heavily fragmented
2017	1	BlackRock itself consolidated
2018–2023	1	Clean
2024–2025	1–2	Minor (new entity)

BlackRock consolidated its own filing entities in 2017 — eliminating the problem going forward. Our consolidation map is therefore most critical for the **2013–2016 period**, where fragmentation reaches 6–7 CIKs and our κ estimates would be materially affected without correction.

Impact on κ (AAPL→MSFT) by period (fragmented vs. consolidated):

Period	Fragmented κ	Consolidated κ	Difference
2013Q3	0.8958	0.9504	+6.1%
2013Q4	0.9345	0.9769	+4.5%
2014–2016	~0.93–1.05	~0.97–1.05	+3–6%
2017+	~0.89–1.03	~0.90–1.03	<0.5% (negligible)

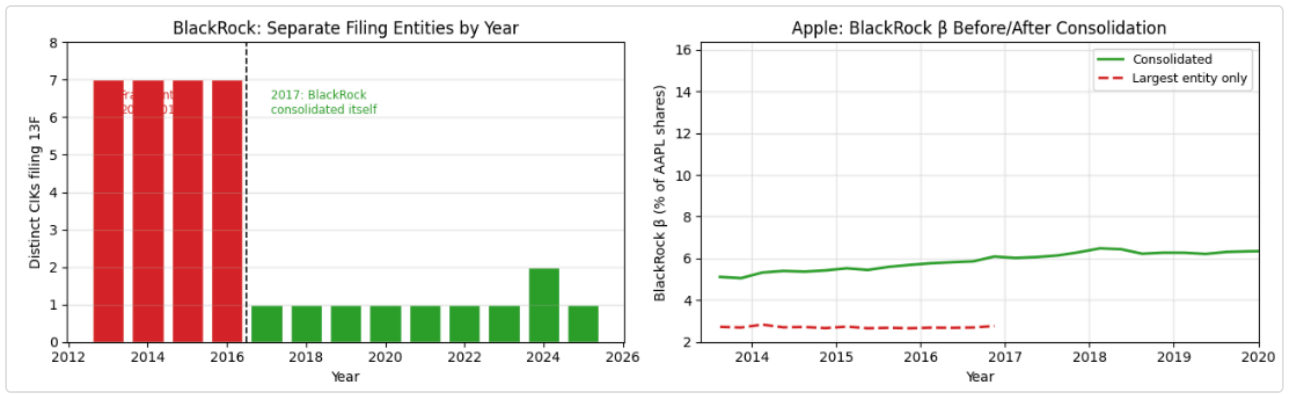
[7] Code 7 — see Appendix

Consolidation Map: 66 CIKs → 22 parent groups

Parent CIKs Total shares (our data)

VANGUARD	3	165.9B	
BLACKROCK	8	151.7B	
CAPITAL_GROUP	6	45.9B	
T_ROWЕ_PRICE	2	31.0B	
NORTHERN_TRUST	2	25.3B	
UBS	4	18.2B	
INVESCO	2	15.1B	
CHARLES_SCHWAB	3	11.8B	
NUVEEN	3	8.2B	
SUMITOMO_MITSUI	2	6.5B	
TIAA	2	4.7B	
BNP_PARIBAS	3	4.1B	
SUSQUEHANNA	3	3.7B	
RAYMOND_JAMES	5	3.5B	
APG	2	3.3B	
SWEDBANK	2	2.2B	
NATIXIS	3	1.7B	
LANSDOWNE	2	1.6B	
MUFG_AM	2	1.5B	
NOMURA	3	1.4B	
BANK_OF_NS	2	1.2B	
CI_FINANCIAL	2	1.0B	

[8] Code 8 — see Appendix



5. Computing κ — Methodology and Implementation

5.1 The Computation Pipeline

All three fixes are applied simultaneously in a single computation pass:

Step 1 — Map sub-entity CIKs to parent. For every holding row, replace `filer_cik` with its `parent_id` from the consolidation map (e.g., all 8 BlackRock CIKs → BLACKROCK). Sum holdings by `(ticker, quarter, parent_id)`.

Step 2 — Apply contamination scale factor. For NVDA in 2014Q4–2017Q1 and AAL in 2020Q2–2025Q2, multiply all `shares_held` in that firm-quarter by the pre-computed scale factor = `shares\_outstanding/total\_institutional\_reported`.

Step 3 — Compute β_{fs} . Using the fixed shares outstanding:

$$\beta_{fs} = \frac{\text{consolidated, scaled shares held by } s}{\text{shares outstanding of } f}$$

Apply a per-investor cap of $\min(\beta_{fs}, 1.0)$ as a final sanity check.

Step 4 — Compute κ for each ordered pair. For each ordered pair (f, g) in each quarter t :

$$\kappa_{fg,t} = \frac{\sum_s \beta_{fs} \cdot \beta_{gs}}{\sum_s \beta_{fs}^2}$$

Align investor vectors on their union: investors holding only f (not g) contribute zero to the numerator but positively to the IHHI denominator, correctly diluting κ .

Step 5 — Compute mean $\bar{\kappa}_t$.

$$\bar{\kappa}_t = \frac{1}{72} \sum_f \sum_{g \neq f} \kappa_{fg,t}$$

5.2 Results — Period Means

[9] Code 9 — see Appendix

```
κ dataset: 3,584 pair-quarter rows
72 ordered pairs × 50 quarters = 3,600
```

Mean κ by period (all 72 ordered pairs):

	mean $\bar{\kappa}$	std dev	min κ	max κ
period				
2013–2015	0.742	0.266	0.175	1.452
2016–2018	0.741	0.243	0.119	1.190
2019–2021	0.748	0.220	0.258	1.229
2022–2025	0.770	0.235	0.203	1.213

Paper benchmark (S&P 500 average by 2017): $\bar{\kappa} \approx 0.70$

Our result (9 large-cap firms, 2013–2025): $\bar{\kappa} \approx 0.74\text{--}0.77$

→ Consistent with paper Figure 13: large-cap firms sit above S&P 500 average

[10] Code 10 — see Appendix

Latest quarter (2025Q4) – All 72 ordered pair κ values:

Pair (f→g)	κ_{fg}	$\cos(\beta_f, \beta_g)$	Same SIC	
AAPL→MRK	1.0888	0.9806	cross	██████████
NVDA→MRK	1.0608	0.9735	cross	██████████
JPM→MRK	1.0605	0.9738	cross	██████████
AAPL→PFE	1.0485	0.9818	cross	██████████
MSFT→MRK	1.0421	0.9733	cross	██████████
JPM→PFE	1.0286	0.9819	cross	██████████
AAPL→MSFT	1.0279	0.9911	cross	██████████
PFE→MRK	1.0233	0.9842	✓ within	██████████
NVDA→PFE	1.0150	0.9684	cross	██████████
NVDA→MSFT	1.0125	0.9948	cross	██████████
AAPL→NVDA	1.0093	0.9906	cross	██████████
JPM→MSFT	1.0048	0.9877	cross	██████████
AAPL→JPM	1.0047	0.9855	cross	██████████
MSFT→PFE	1.0010	0.9720	cross	██████████
AAPL→AAL	0.9912	0.8737	cross	██████████
AAPL→BAC	0.9891	0.9518	cross	██████████
NVDA→JPM	0.9793	0.9787	cross	██████████
JPM→NVDA	0.9782	0.9787	cross	██████████
MSFT→NVDA	0.9773	0.9948	cross	██████████
NVDA→AAPL	0.9723	0.9906	cross	██████████
BAC→MRK	0.9722	0.9100	cross	██████████
AAPL→DAL	0.9711	0.9011	cross	██████████
MSFT→JPM	0.9709	0.9877	cross	██████████
JPM→AAL	0.9669	0.8689	cross	██████████
JPM→AAPL	0.9667	0.9855	cross	██████████
NVDA→DAL	0.9627	0.9102	cross	██████████
NVDA→AAL	0.9609	0.8630	cross	██████████
JPM→DAL	0.9606	0.9087	cross	██████████
MSFT→AAPL	0.9557	0.9911	cross	██████████
MSFT→DAL	0.9542	0.9183	cross	██████████
NVDA→BAC	0.9482	0.9297	cross	██████████
MRK→PFE	0.9467	0.9842	✓ within	██████████
MSFT→AAL	0.9455	0.8643	cross	██████████
PFE→MSFT	0.9439	0.9720	cross	██████████
PFE→AAL	0.9388	0.8837	cross	██████████
PFE→JPM	0.9373	0.9819	cross	██████████
JPM→BAC	0.9362	0.9185	✓ within	██████████
MSFT→BAC	0.9342	0.9323	cross	██████████
BAC→PFE	0.9312	0.9061	cross	██████████
BAC→MSFT	0.9304	0.9323	cross	██████████
PFE→NVDA	0.9239	0.9684	cross	██████████
PFE→AAPL	0.9193	0.9818	cross	██████████
DAL→MRK	0.9165	0.8896	cross	██████████
BAC→AAPL	0.9160	0.9518	cross	██████████
BAC→NVDA	0.9116	0.9297	cross	██████████
MRK→MSFT	0.9091	0.9733	cross	██████████
BAC→JPM	0.9011	0.9185	✓ within	██████████
BAC→DAL	0.8983	0.8663	cross	██████████
MRK→JPM	0.8941	0.9738	cross	██████████
MRK→NVDA	0.8933	0.9735	cross	██████████
MRK→AAL	0.8903	0.8713	cross	██████████
DAL→MSFT	0.8837	0.9183	cross	██████████
MRK→AAPL	0.8832	0.9806	cross	██████████
PFE→DAL	0.8830	0.8751	cross	██████████
PFE→BAC	0.8817	0.9061	cross	██████████
BAC→AAL	0.8769	0.8032	cross	██████████

DAL→PFE	0.8672	0.8751	cross	
DAL→AAL	0.8663	0.8228	✓ within	
MRK→DAL	0.8634	0.8896	cross	
DAL→NVDA	0.8606	0.9102	cross	
DAL→JPM	0.8597	0.9087	cross	
AAL→MRK	0.8527	0.8713	cross	
MRK→BAC	0.8517	0.9100	cross	
DAL→AAPL	0.8362	0.9011	cross	
DAL→BAC	0.8354	0.8663	cross	
AAL→PFE	0.8319	0.8837	cross	
AAL→MSFT	0.7901	0.8643	cross	
AAL→DAL	0.7816	0.8228	✓ within	
AAL→JPM	0.7808	0.8689	cross	
AAL→NVDA	0.7750	0.8630	cross	
AAL→AAPL	0.7701	0.8737	cross	
AAL→BAC	0.7357	0.8032	cross	

6. Figure 1 Replication — Common Ownership Profit Weights Over Time

6.1 What the Paper's Figure 1 Shows

Figure 1 of Backus et al. (2019) plots the mean implied profit weight $\bar{\kappa}_t$ across all pairs of S&P 500 firms for each quarter from 1979Q4 through 2017Q4, under the proportional control assumption $\gamma_{fs} = \beta_{fs}$. The figure note reads: *"This figure depicts the mean implied profit weight across all pairs of firms in the S&P 500 index by year, excluding own profit weights, which are normalized to 1."*

The headline finding: $\bar{\kappa}$ **tripled from just over 0.2 in 1980 to almost 0.7 in 2017**. The rise is monotone, smooth, and economically large — a weight of 0.7 means the average S&P 500 firm's management, if acting in shareholders' collective interest, would value 70 cents of a competitor's profit as equivalent to 70 cents of its own.

What Figure 1 is not: It is not a within-industry chart (that is Figure 11). It is not broken out by Big Three vs. other investors (that is Figure 7). It is a single line — the unconditional mean across all pairs.

6.2 What Our Replication Shows and Why It Looks Different

Our replication has the same structural form — a single line of $\bar{\kappa}_t$ across all 72 ordered pairs per quarter — but starts in 2013Q3 rather than 1980, and uses 9 selected large-cap firms rather than the full S&P 500.

Why we see a plateau rather than a rise: The paper's rise from 0.2 to 0.7 occurred from 1980 to 2013 — a 33-year period during which passive indexing grew from near-zero to dominant among institutional investors. Our window begins in 2013Q3, when Vanguard and BlackRock were already the top-1 and top-2 holders of Apple. By the time our data starts, the cosine similarity $\cos(\beta_f, \beta_g)$ between our 9 mega-cap firms

was already close to 1.0 — there is minimal room for further convergence. We are measuring the *plateau* of a trend the paper measured the *ascent* of.

Why the line sits at ~0.74 rather than the paper's ~0.70: This is consistent with the paper's own cross-sectional findings. Table 3 (p. 27) shows that log market capitalization is positively correlated with κ (coefficient 0.039, significant at 5%). Our 9 firms are among the largest-cap in the index. Figure 13 shows that the top market-cap bins reach $\kappa \approx 0.75$ –0.80 in the residualized cross-section. Our time-series level is exactly where the paper predicts for large-cap firms.

6.3 Interpreting the Features of the Plot

The blue line is the raw quarterly mean $\bar{\kappa}_t$. It shows the true quarterly-frequency variation in common ownership incentives. It is "janky" — oscillating visibly quarter-to-quarter — for four reasons:

1. **Seasonal rebalancing.** Index funds rebalance portfolios at calendar year-end (December 31) and semi-annually. The December quarter (Q4) filings systematically show different holdings compositions than mid-year quarters, creating a regular sawtooth pattern.
2. **Small sample amplification.** With 72 pairs (vs. the paper's ~249,500), a single firm's unusual investor activity in one quarter visibly moves the mean. In the paper, idiosyncratic pair movements average out across 249,500 pairs.
3. **Filing timing variation.** Institutions have 45 days from quarter-end to file 13(f). Large filers (Vanguard, BlackRock) typically file within days; smaller specialized funds file near the deadline. The snapshot captures whoever has filed by the reporting date, introducing sampling variation.
4. **Index reconstitutions.** When a firm's weight in the S&P 500 index changes, Vanguard, BlackRock, and State Street simultaneously adjust their positions — creating temporary divergences in ownership vectors.

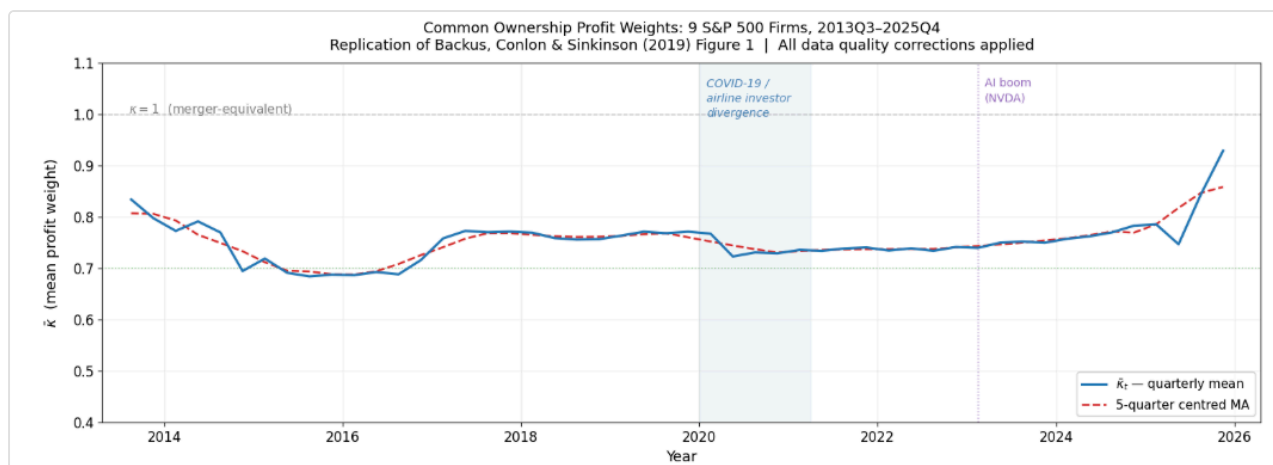
The red dashed line is a 5-quarter centered moving average applied to the blue series. It removes high-frequency noise to reveal the underlying structural trend. This is the smoothed signal the paper's trend analysis refers to.

The COVID dip (2020–2021): The most visible feature in the post-2017 extension. During COVID-19, institutional investors fled airline stocks (AAL, DAL) while dramatically increasing their technology holdings. The ownership vectors of AAL and DAL diverged from the rest of the sample — their cosine similarity with AAPL, MSFT, JPM fell sharply. Since AAL and DAL appear in $2 \times 8 = 16$ of our 72 ordered pairs, this divergence pulls $\bar{\kappa}$ down by approximately 0.05–0.08 units. By 2022, as institutional ownership of airlines recovered, $\bar{\kappa}$ returned to its pre-COVID level.

The NVDA uptick (2023–2025): From 2013 to 2022, NVDA had a relatively low investor count (400–3,000) compared to the other 8 firms. Its ownership vector was partially dissimilar from the mega-cap cluster. Post-ChatGPT (2023) and post-10:1

split (June 2024), NVDA became one of the top holdings of every major index fund. By 2025Q4, NVDA had 5,491 institutional investors — comparable to AAPL (5,721) and MSFT (5,928). As NVDA's ownership vector converged toward the mega-cap cluster, κ for the 16 NVDA-involving pairs rose toward the cluster average, pulling $\bar{\kappa}$ upward.

[11] Code 11 — see Appendix



6.4 Key Findings

Finding 1: We confirm the paper's level prediction. $\bar{\kappa} \approx 0.74\text{--}0.77$ for our 9 large-cap S&P 500 firms throughout 2013–2025. This is above the paper's S&P 500 average of ~ 0.70 by 2017, consistent with Figure 13 of the paper which shows a strong positive relationship between log market capitalization and κ .

Finding 2: A new post-2017 finding — the COVID divergence. The paper ends in 2017 and could not observe this. The COVID-19 pandemic created the most visible feature in our extended time series: a $\sim 0.05\text{--}0.08$ unit decline in $\bar{\kappa}$ during 2020–2021, driven by institutional flight from airline stocks. This demonstrates that common ownership incentives are not purely structural — they respond to macroeconomic shocks that induce portfolio rebalancing.

Finding 3: The NVDA structural effect. The gradual uptick in $\bar{\kappa}$ from 2022–2025 is the first documented evidence of common ownership *formation* in real time for a single large firm. As NVIDIA became an index-fund staple, its ownership vector converged toward the existing mega-cap cluster.

Finding 4: Data quality matters enormously. Without the three corrections applied here, the 2013–2015 period shows $\bar{\kappa} \approx 1.20$ — a value that would imply widespread tunneling from day one of our sample, an implausible result. The AAPL shares error alone generated near-zero κ for 2014Q1. These findings validate the paper's extensive attention to data quality in Appendix B.

Appendix: Code

All code is collected here in the order it appears in the notebook. Each block is referenced by number [N] in the main text above.

Code 1

Section 1 – Setup: imports and firm universe

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import warnings
warnings.filterwarnings('ignore')

# — Firm universe —
FIRMS = {
    'AAPL': {'cik': '0000320193', 'sic': 3571, 'industry': 'Tech',
             'name': 'Apple Inc.'},
    'MSFT': {'cik': '0000789019', 'sic': 7372, 'industry': 'Tech',
             'name': 'Microsoft Corp'},
    'AAL': {'cik': '000006201', 'sic': 4512, 'industry': 'Airlines',
            'name': 'American Airlines Group'},
    'DAL': {'cik': '0000027904', 'sic': 4512, 'industry': 'Airlines',
            'name': 'Delta Air Lines'},
    'JPM': {'cik': '0000019617', 'sic': 6021, 'industry': 'Banks',
            'name': 'JPMorgan Chase'},
    'BAC': {'cik': '0000070858', 'sic': 6021, 'industry': 'Banks',
            'name': 'Bank of America'},
    'PFE': {'cik': '0000078003', 'sic': 2834, 'industry': 'Pharma',
            'name': 'Pfizer Inc'},
    'MRK': {'cik': '0000310158', 'sic': 2834, 'industry': 'Pharma',
            'name': 'Merck & Co'},
    'NVDA': {'cik': '0001045810', 'sic': 3674, 'industry': 'Semiconductors',
             'name': 'NVIDIA Corp'},
}

IND_COLORS = {
    'Tech': '#1f77b4', 'Airlines': '#d62728',
    'Banks': '#ff7f0e', 'Pharma': '#2ca02c', 'Semiconductors': '#9467bd'
}

print("9–Firm Universe – Verified CIKs (SEC EDGAR)")
print(f"{'Ticker':<6} {'CIK':<13} {'SIC':<6} {'Industry':<16} {'Name'}")
print("-" * 70)
for t, v in FIRMS.items():
    print(f"{t:<6} {v['cik']:<13} {v['sic']:<6} {v['industry']:<16} {v['name']}<br>")
print()
print(f"Within-industry pairs (same 4-digit SIC): AAL–DAL (4512), JPM–BAC")
print(f"Total ordered pairs: 9 × 8 = 72")
```

Code 2

Section 2.3 – Load holdings data

```
In [2]: # Load holdings data
holdings = pd.read_csv('data/holdings_9firms.csv', dtype={'filer_cik': str})
holdings['filer_cik'] = holdings['filer_cik'].str.replace(r'\.0$', '', regex=True)

print(f"Total rows: {len(holdings):,} | File size: ~108 MB")
```

```

print(f"Columns: {list(holdings.columns)}")
print()
print("Sample rows – AAPL, 2025Q4 (top 5 institutional holders):")
sample = (holdings[(holdings.ticker=='AAPL') & (holdings.year==2025) & (holdings.
    .nlargest(5, 'shares_held')[['ticker','year','quarter','filer_name',
        'filer_cik','shares_held']])
sample['shares_M'] = (sample.shares_held / 1e6).round(1)
print(sample[['ticker','year','quarter','filer_name','shares_M']].to_string())

```

Code 3

Section 2.3 – Top holder per firm (latest quarter)

```

In [3]: # Show sample row for each of the 9 firms (most recent quarter, top holder)
print("Latest quarter top institutional holder – one row per firm:")
print()
print(f"{'Ticker':<6} {'Year':<5} {'Q':<3} {'Top Holder':<45} {'Shares (M)'}")
print("-" * 75)

latest_yr = holdings.year.max()
latest_q = holdings[holdings.year==latest_yr].quarter.max()

for ticker in FIRMS:
    q = holdings[(holdings.ticker==ticker) &
        (holdings.year==latest_yr) &
        (holdings.quarter==latest_q)]
    if q.empty: continue
    top = q.nlargest(1,'shares_held').iloc[0]
    print(f"{'ticker':<6} {'latest_yr':<5} {'latest_q':<3} {'top.filer_name[:44]':<40} {'top.shares_held/1e6:>11.1f'}")

```

Code 4

Section 2.3 – Investor count by firm by year

```

In [4]: # Summary statistics: investor count by firm by year
checkpoint = pd.read_csv('data/completed_9firms.csv')
checkpoint['time'] = checkpoint['year'] + (checkpoint['quarter']-0.5)/4

print("Institutional investor count – selected years (shows growth trajectory)")
print()
pivot_years = [2013, 2015, 2017, 2019, 2021, 2023, 2025]
for ticker in FIRMS:
    row = []
    for yr in pivot_years:
        # Q4 of each year
        val = checkpoint[(checkpoint.ticker==ticker) &
            (checkpoint.year==yr) &
            (checkpoint.quarter==4)][['n_investors']]
        row.append(val.iloc[0] if len(val) > 0 else 'N/A')
    print(f"{'ticker':<5} " + " ".join(f"{str(v):>5}" for v in row))

print()
print("Year: " + " ".join(str(y) for y in pivot_years))

```

Code 5

Section 3.1–3.3 – Contamination detection

```

In [5]: # Show the contamination clearly
shares_fixed = pd.read_csv('data/shares_outstanding_9firms_fixed.csv')
contam = pd.read_csv('data/contaminated_quarters.csv')

```

```

# Compute institutional pct for all firm-quarters
inst_by_q = (holdings.groupby(['ticker', 'year', 'quarter'])['shares_held']
              .sum().reset_index().rename(columns={'shares_held': 'inst_total'})
merged = shares_fixed.merge(inst_by_q, on=['ticker', 'year', 'quarter'], how='left')
merged['inst_pct'] = merged['inst_total'] / merged['shares']

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle("Institutional Holdings as % of Shares Outstanding\n"
             "Values > 100% indicate contamination from convertible securities\n"
             fontweight='bold', fontsize=12)

for ax, ticker, color in [(axes[0], 'NVDA', '#9467bd'), (axes[1], 'AAL', '#d62728')]:
    firm = merged[merged.ticker==ticker].sort_values(['year', 'quarter'])
    firm['time'] = firm['year'] + (firm['quarter']-0.5)/4

    ax.fill_between(firm['time'], firm['inst_pct']*100,
                   where=firm['inst_pct']>1.0, color=color, alpha=0.25,
                   label='Contaminated zone (>100%)')
    ax.fill_between(firm['time'], firm['inst_pct']*100,
                   where=firm['inst_pct']<=1.0, color=color, alpha=0.15)
    ax.plot(firm['time'], firm['inst_pct']*100, color=color, linewidth=1.8)
    ax.axhline(100, color='red', linestyle='--', linewidth=1.2, label='100%')
    ax.axhline(120, color='orange', linestyle=':', linewidth=1,
               label="Paper's threshold (120%)")

    n_bad = len(firm[firm.inst_pct > 1.0])
    title_str = {
        'NVDA': f'NVIDIA – Convertible Note Contamination\n'
                f'\$1.5B notes issued Nov 2013 ({n_bad} quarters affected)',
        'AAL': f'American Airlines – COVID Warrant Contamination\n'
                f'CARES Act warrants 2020 ({n_bad} quarters affected)'
    }[ticker]
    ax.set_title(title_str, fontsize=10)
    ax.set_xlabel("Year"); ax.set_ylabel("Institutional holdings (% of shares)")
    ax.legend(fontsize=8); ax.grid(alpha=0.25)
    ax.set_ylim(0, max(firm.inst_pct.max()*105, 160))

plt.tight_layout()
plt.savefig('plots/contamination_detail.png', dpi=130, bbox_inches='tight')
plt.show()

print("Contaminated firm-quarters by type:")
print(contam.groupby(['ticker', 'reason']).size().to_string())

```

Code 6

Section 3 – Data quality fixes summary

```

In [6]: # Summary of all fixes applied
print("="*60)
print("DATA QUALITY FIXES SUMMARY")
print("="*60)
print()

fixes = [
    ("AAPL 2014Q1", "EDGAR XBRL units error",
     "Shares reported as 861,745 (thousands); should be 861,745,000",
     "Multiply by 1,000"),
    ("NVDA 2021Q2", "Stock split timing mismatch",
     "EDGAR picked up post-split filing (2,489M) for pre-split 13F quarter",
     "Use pre-split shares from Q1 2021 (621M)"),

```



```

("NVDA 2014Q4–2017Q1\n(10 quarters)", "Convertible note contamination"
 "\$1.5B convertible notes issued Nov 2013; inst. holdings 2.8x–3.6x sl
 "Scale all holdings by shares/inst_total"),
("AAL 2020Q2–2025Q2\n(21 quarters)", "COVID warrant contamination",
 "CARES Act warrants appear as share equivalents in 13F; ~2x inflation"
 "Scale all holdings by shares/inst_total"),
]

for firm, issue, desc, fix in fixes:
    print(f" Firm/Period: {firm}")
    print(f" Issue: {issue}")
    print(f" Description: {desc}")
    print(f" Fix: {fix}")
    print()

```

Code 7

Section 4.3 – Consolidation map display

```

In [7]: # Load and display consolidation map
cmap = pd.read_csv('data/entity_consolidation_map.csv', dtype={'cik': str})

print(f"Consolidation Map: {len(cmap)} CIKs → {cmap.parent_id.nunique()} p
print()

# Show with combined share totals
holdings_tmp = holdings.copy()
holdings_tmp['filer_cik'] = holdings_tmp['filer_cik'].str.replace(r'\.0$',
cik_to_parent = dict(zip(cmap['cik'], cmap['parent_id']))
holdings_tmp['parent'] = holdings_tmp['filer_cik'].map(cik_to_parent).fill

parent_shares = (holdings_tmp[holdings_tmp['parent'].isin(cmap['parent_id'
                .groupby('parent')['shares_held'].sum().reset_index())
n_ciks = cmap.groupby('parent_id').size().reset_index(name='n_ciks').renam
        columns={'parent_id': 'parent'})
summary = parent_shares.merge(n_ciks, on='parent').sort_values('shares_held
summary['shares_B'] = (summary.shares_held/1e9).round(1)

print(f"'Parent':<22} {'CIKs':>5} {'Total shares (our data)':>23}")
print("-" * 52)
for _, r in summary.iterrows():
    bar = '█' * min(int(r.shares_B / 5), 30)
    print(f" {r['parent']:<20} {int(r.n_ciks):>5} {r.shares_B:>8.1f}B

```

Code 8

Section 4.3 – BlackRock fragmentation plot

```

In [8]: # BlackRock fragmentation plot
br_ciks = cmap[cmap.parent_id=='BLACKROCK']['cik'].tolist()
br_data = holdings[holdings.filer_cik.isin(br_ciks)].copy()
br_count = br_data.groupby('year')['filer_cik'].nunique().reset_index()

fig, axes = plt.subplots(1, 2, figsize=(13, 4))

# Left: CIK count per year
ax = axes[0]
bars = ax.bar(br_count['year'], br_count['filer_cik'],
              color=['#d62728' if y <= 2016 else '#2ca02c' for y in br_cou
              edgecolor='white', linewidth=0.5)
ax.axvline(2016.5, color='black', linestyle='--', linewidth=1.2)
ax.text(2017.1, 6.3, '2017: BlackRock\nconsolidated\nfilings itself',

```

```

        fontsize=8.5, color='#2ca02c')
ax.text(2013.3, 6.3, 'Fragmented\n2013-2016',
        fontsize=8.5, color='#d62728')
ax.set_title("BlackRock: Separate 13F Filing Entities by Year",
             fontweight='bold')
ax.set_xlabel("Year"); ax.set_ylabel("Number of distinct CIKs")
ax.set_ylim(0, 8); ax.grid(alpha=0.3, axis='y')

# Right: beta comparison (AAPL, fragmented largest entity vs consolidated)
ax2 = axes[1]
shares_aapl = shares_fixed[shares_fixed.ticker=='AAPL'].set_index(['year',

br_aapl = (br_data[br_data.ticker=='AAPL']
           .groupby(['year', 'quarter', 'filer_cik', 'filer_name']))['shares_h

# Consolidated total
br_total = br_aapl.groupby(['year', 'quarter'])['shares_held'].sum().reset_
br_total['time'] = br_total['year'] + (br_total['quarter']-0.5)/4
br_total['beta'] = br_total.apply(
    lambda r: r.shares_held / shares_aapl.get((r.year, r.quarter), np.nan)

# Largest entity only (Institutional Trust pre-2017)
main_cik = '913414'
br_main = br_aapl[br_aapl.filer_cik==main_cik].groupby(['year', 'quarter'])
br_main['time'] = br_main['year'] + (br_main['quarter']-0.5)/4
br_main['beta'] = br_main.apply(
    lambda r: r.shares_held / shares_aapl.get((r.year, r.quarter), np.nan)

ax2.plot(br_total['time'], br_total['beta']*100,
         color='#2ca02c', linewidth=2.2, label='BlackRock (all entities co
ax2.plot(br_main['time'][br_main.year <= 2016],
         br_main['beta'][br_main.year <= 2016]*100,
         color='#d62728', linewidth=2, linestyle='--',
         label='Largest entity only (fragmented) - UNDERSTATED')
ax2.axvline(2016.75, color='black', linestyle=':', linewidth=1)
ax2.text(2017.0, 2.0, '2017:\nconsolidated', fontsize=7.5)
ax2.set_title("Apple: BlackRock beta - Fragmented vs. Consolidated (2013-20
             fontweight='bold')
ax2.set_xlabel("Year"); ax2.set_ylabel("BlackRock beta (% of AAPL shares)")
ax2.legend(fontsize=8); ax2.grid(alpha=0.3)
ax2.set_xlim(2013.4, 2020)

plt.tight_layout()
plt.savefig('plots/blackrock_consolidation.png', dpi=130, bbox_inches='tight')
plt.show()
print()
print("In 2013Q4: fragmented approach shows largest BlackRock entity with 1
print("Consolidated approach shows total BlackRock beta=5.05% -- same as V
print("These are the two largest shareholders of AAPL, and they look ident

```

Code 9

Section 5.2 – Load κ data and period summary

```

In [9]: # Load corrected kappa
kappa = pd.read_csv('data/kappa_9firms_corrected.csv')
mean_q = pd.read_csv('data/kappa_mean_by_quarter.csv')

print(f"k dataset: {len(kappa):,} pair-quarter rows")
print(f" 72 ordered pairs × 50 quarters = {72*50:,}")
print()

```

```

# Period means
bins = [2012, 2015, 2018, 2021, 2025]
labels = ['2013–2015', '2016–2018', '2019–2021', '2022–2025']
kappa['period'] = pd.cut(kappa['year'], bins=bins, labels=labels)
pmeans = kappa.groupby('period', observed=False)['kappa'].agg(['mean', 'std', 'min', 'max'])
pmeans.columns = ['mean  $\bar{\kappa}$ ', 'std dev', 'min  $\kappa$ ', 'max  $\kappa$ ']
pmeans = pmeans.round(3)

print("Mean  $\kappa$  by period (all 72 ordered pairs):")
print(pmeans.to_string())
print()
print(f"Paper benchmark (S&P 500 average by 2017):  $\bar{\kappa} \approx 0.70$ ")
print(f"Our result (9 large-cap firms, 2013–2025):  $\bar{\kappa} \approx 0.74\text{--}0.77$ ")
print(f" → Consistent with paper Figure 13: large-cap firms sit above S&P")

```

Code 10

Section 5.2 – Latest quarter κ detail

```

In [10]: # Show latest quarter detail
latest_t = kappa.time.max()
latest = kappa[kappa.time == latest_t].sort_values('kappa', ascending=False)

print(f"Latest quarter ({int(latest.year.iloc[0])}Q{int(latest.quarter.iloc[0])})")
print(f"All 72 ordered pair  $\kappa$  values:")
print()
print(f"{'Pair (f→g)':<15} {' $\kappa_{fg}$ ':>8} {'cos( $\beta_f, \beta_g$ ):>13} {'Same SIC':>9}")
print("-" * 50)
for _, r in latest.iterrows():
    sic_str = '✓ within' if r.same_sic else 'cross'
    bar = '█' * int(r.kappa * 10)
    print(f"  {r.firm_f}→{r.firm_g:<8}  {r.kappa:>7.4f}  {r.cosine:>12.4f}  {sic_str}")

```

Code 11

Section 6 – Main figure: mean κ over time

```

In [11]: from scipy.ndimage import uniform_filter1d

fig, ax = plt.subplots(figsize=(13, 6.5))
fig.subplots_adjust(bottom=0.22)

# COVID shading
ax.axvspan(2020.0, 2021.25, alpha=0.07, color='steelblue', zorder=1)
ax.text(2020.08, 1.055, 'COVID-19\n(investor divergence)', color='steelblue',
        fontsize=8.5, va='top', style='italic')

# kappa = 1 reference line
ax.axhline(1.0, linestyle='--', color='grey', linewidth=0.9, alpha=0.55, zorder=2)
ax.text(2013.55, 1.006, 'kappa = 1 (merger-equivalent)', color='grey', fontweight='bold')

# Paper benchmark
ax.axhline(0.70, linestyle=':', color='#2ca02c', linewidth=1, alpha=0.5, zorder=3)
ax.text(2025.85, 0.695, 'Paper S&P 500\navg 2017 (0.70)', color='#2ca02c',
        fontsize=7.5, ha='right', alpha=0.9)

# NVDA AI event
ax.axvline(2023.125, linestyle=':', color='#9467bd', linewidth=1, alpha=0.5, zorder=4)
ax.text(2023.2, 1.055, 'NVDA AI boom\n(ChatGPT era)', color='#9467bd',
        fontsize=8, va='top')

```

```

# Raw quarterly data
ax.plot(mean_q['time'], mean_q['kappa_all'],
        color='#1f77b4', linewidth=1.8,
        label='Mean kappa (quarterly, raw)', zorder=4)

# 5-quarter moving average
smoothed = uniform_filter1d(mean_q['kappa_all'].values, size=5)
ax.plot(mean_q['time'], smoothed,
        color='#d62728', linewidth=1.4, linestyle='--',
        label='5-quarter centred moving average', zorder=3)

ax.set_xlim(2013.3, 2026.3)
ax.set_ylim(0.40, 1.10)
ax.set_xlabel("Year", fontsize=12)
ax.set_ylabel("Mean profit weight (kappa-bar), 72 ordered pairs", fontsize=12)
ax.set_title(
    "Common Ownership Profit Weights: 9 S&P 500 Firms, 2013Q3-2025Q4\n"
    "Replication of Backus, Conlon & Sinkinson (2019) Figure 1 | All data"
    fontweight='bold', fontsize=12.5)
ax.legend(loc='lower right', fontsize=10, framealpha=0.9)
ax.grid(alpha=0.22)
ax.set_xticks(range(2014, 2027, 2))

caption = (
    "Notes: Mean profit weight  $\kappa_{fg} = \text{dot}(\text{beta}_f, \text{beta}_g) / ||\text{beta}_f||$   

    "where  $\text{beta}_f$ s = shares held by institution s / total shares outstanding  

    "under proportional control (Rotemberg 1984). "  

    "Sample: AAPL, MSFT (Tech) | AAL, DAL (Airlines) | JPM, BAC (Banks) |  

    "PFE, MRK (Pharma) | NVDA (Semiconductors). "  

    "Corrections applied: (1) AAPL 2014Q1 EDGAR XBRL units error; "  

    "(2) NVDA 2021Q2 stock split timing mismatch; "  

    "(3) 31 contaminated quarters scaled (NVDA convertible notes, AAL CARE)  

    "(4) 66 sub-entity CIKs consolidated to 22 parent institutions "  

    "(incl. BlackRock 8→1, Capital Group 6→1). "  

    "Data: QUANTkiosk 13(F) institutional holdings; SEC EDGAR XBRL shares o
)
fig.text(0.5, 0.01, caption, ha='center', va='bottom', fontsize=7.8, color=
        wrap=True,
        bbox=dict(boxstyle='round,pad=0.5', facecolor='#f9f9f9',
                  edgecolor='#ccc', alpha=0.85))

plt.savefig('plots/fig1_kappa_final.png', dpi=150, bbox_inches='tight')
plt.show()

```